AI AGENTS EXPLAINED

What makes an AI agent deep, not shallow?

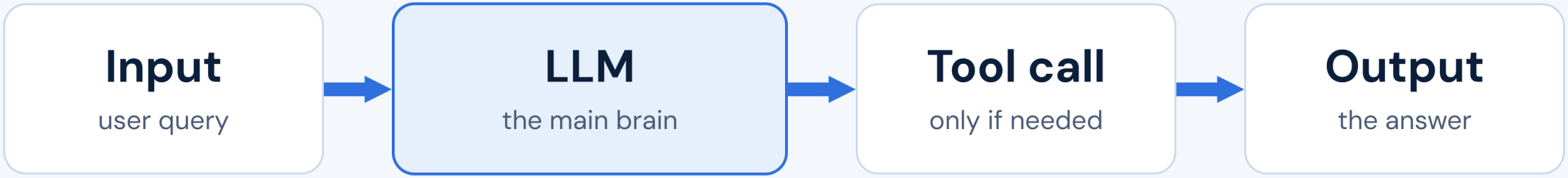
Shallow agent vs deep agent, explained with diagrams.

Shallow vs **Deep**

Shahab Khan

01 · SHALLOW AGENT

Shallow agent: the simple pass



No explicit plan

It decides one step at a time.

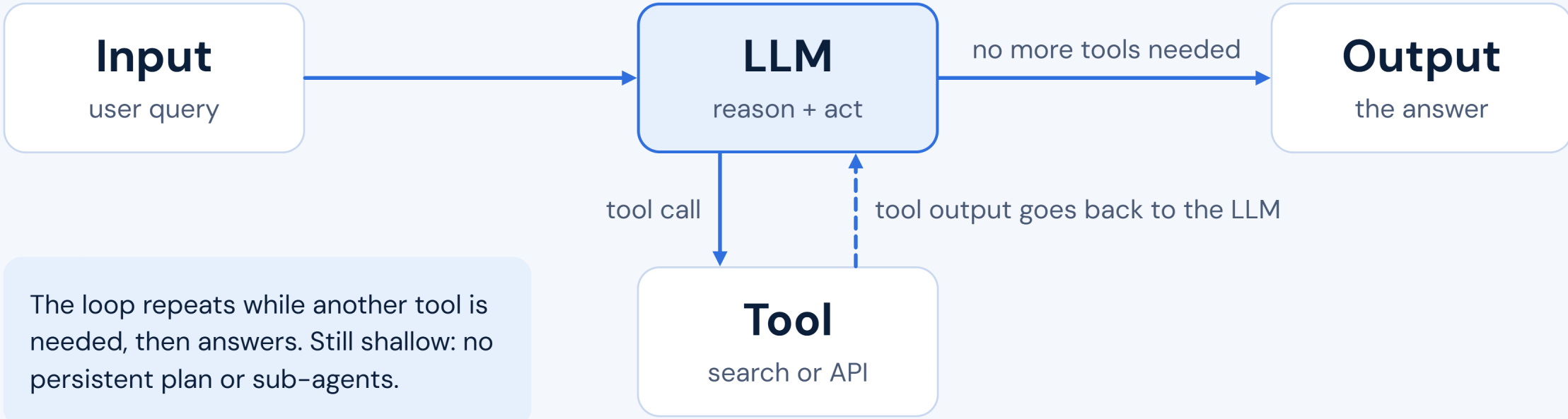
Struggles on long tasks

Complex, multi-step jobs make it lose track.

State lives in context

Everything it knows must fit in one window.

ReAct agent: reason, act, repeat



The loop repeats while another tool is needed, then answers. Still shallow: no persistent plan or sub-agents.

Deep agent: four building blocks

The pattern behind coding agents such as Claude Code.

Planning tool

The agent writes and updates a to-do list.

Sub-agents

Specialist workers with their own context.

File system

Persistent memory the agents read and write.

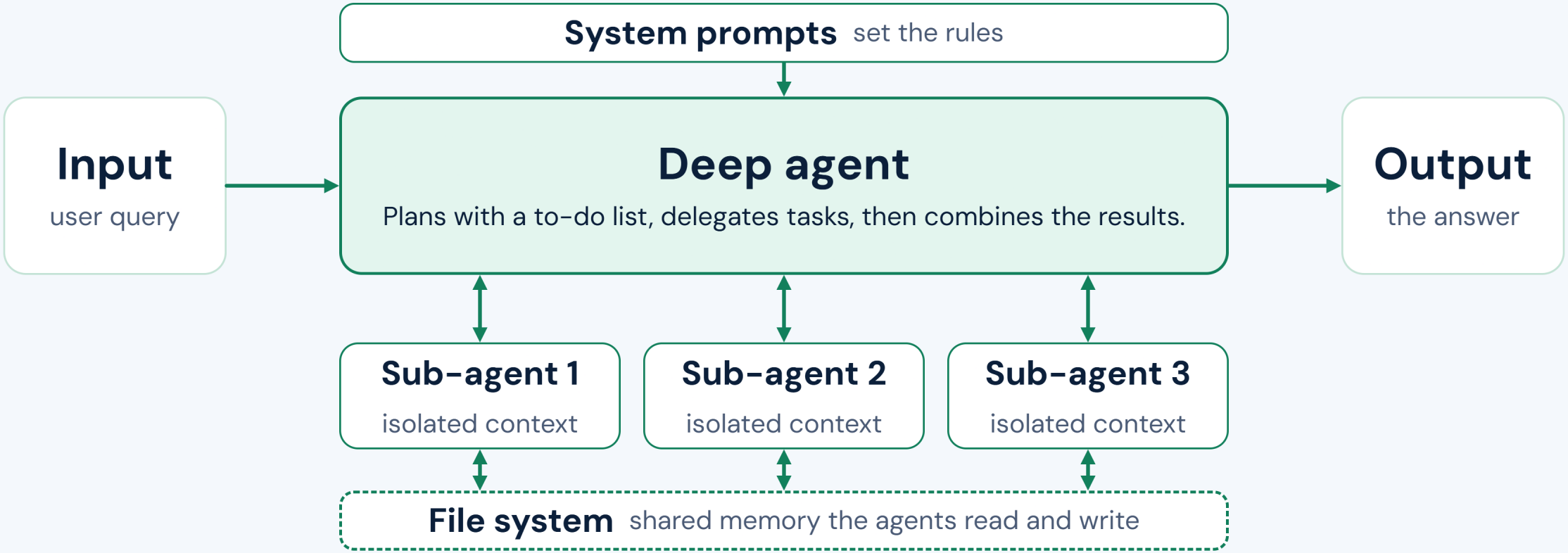
System prompts

Detailed rules for planning, tools and sub-agents.

Same kind of LLM. The architecture around it makes the difference.

Sources: NVIDIA glossary; “Why Most AI Agents Are Shallow”

Deep agent: how the parts connect



Sources: NVIDIA glossary; “Why Most AI Agents Are Shallow”

04 · EXAMPLE (ILLUSTRATIVE)

Example: planning a Paris trip

USER QUERY

"I want to go to Paris for 3 days and 4 nights, and I have $100,000."

To-do list the agent writes

Task 1: Day 1 in Paris	Sub-agent 1
Task 2: Day 2 in Paris	Sub-agent 2
Task 3: Day 3 in Paris	Sub-agent 3
Task 4: Hotels and budget	Sub-agent 4

Sub-agents work, saving to the file system → **Deep agent combines the results** → **One complete trip plan**

05 · COMPARE

Shallow vs deep, side by side

Feature	Shallow agent	Deep agent
Planning	No explicit plan	To-do list it updates
Sub-agents	None	Specialist workers
Memory	Mostly the context	File system
Complex tasks	Struggles on long ones	Breaks them into tasks
Best for	Simple, clear tasks	Long, multi-step work
Trade-off	Simpler to build	Needs sandboxing

Sources: NVIDIA glossary; “Why Most AI Agents Are Shallow”

THE ONE-LINE VERSION

A shallow agent reacts step by step. A deep agent plans, delegates and remembers.

Same kind of LLM. The difference is the architecture around it.

SOURCES AND NOTES

Where this comes from

NVIDIA: What Are Deep Agents?

nvidia.com/en-us/glossary/deep-agents

Deep Agents: Why Most AI Agents Are Shallow

yashmaheshwari.is-a.dev/blog/deep-agents

Good to know

Shallow vs deep is a teaching frame, not a formal standard. Real agents sit on a spectrum, and the Paris example is illustrative.

YOUR TURN

Where did the simple loop stop working for you?

Context limits, long tasks or parallel work? Tell me in the comments.

Follow Shahab Khan for more AI/ML engineering breakdowns